

Ankita Negi

Computational Physicist | Scientific Computing | Numerical Modelling

PhD (Dr. rer. nat.) physicist with expertise in computational science, mathematical modelling, inverse problems, and scientific software development. Experienced in designing scalable Python algorithms, developing computational imaging methods, and implementing numerical optimization techniques. During my research career, I thrived in both independent and collaborative settings and have worked in various multidisciplinary, multi-team projects. I am highly motivated to leverage my modeling and analytical skills to contribute to innovation with real world impact, specifically in climate and energy research.

Skills

Programming languages : Python, MATLAB, Mathematica, Julia , SQL, R

Scientific Computing: Numerical Optimization, Mathematical Modelling, Inverse Problems, Regularization, noise modeling, PDE-based Modelling , Simulation, Computational Imaging, Signal Processing

Software engineering: Git, pytest, Linux, conda, pixi, HPC, SLURM, modular Python Package Development

Data Visualization: Matplotlib, PyVista, Mayavi, Napari

Data Science: Pytorch, scikit-learn, regression, classification, principle component analysis, multivariate statistics

Languages: English (C1), German (B1), Hindi (Native)

Experience

Center for Ultrafast Imaging, Postdoctoral Researcher

Hamburg, Germany

June 2024 – Dec 2025

- Led and coordinated computational research activities in experimental physics, including planning of analysis workflows and alignment of simulation and measurement strategies across research groups.
- Managed interdisciplinary collaboration across experimental and theoretical teams, supporting data interpretation and scientific communication in international projects.
- Contributed to research design for studies of material response under coupled X-ray and optical excitation in advanced spectroscopy experiments.

Deutsches Elektronen-Synchrotron (DESY), Doctoral Researcher

Hamburg, Germany

Oct 2020 – Apr 2024

- Conducted doctoral research in computational and experimental physics focusing on inverse problems and numerical modeling in X-ray optics and nuclear resonance spectroscopy.
- Investigated and applied optimization and statistical methods for analysis of high-dimensional experimental datasets in collaboration with applied mathematicians.
- Presented results in international collaborations and contributed to scientific discussions and publications.

University of Cologne (CRC 183), Student Assistant

Cologne, Germany

Oct 2018 – Jan 2019

- Supported research on quantum and topological systems within a collaborative condensed matter physics program.
- Ran numerical simulations of quantum models using HPC resources and differential equation solvers.
- Assisted in teaching computational physics to bachelor students.

Indian Institute of Science Education and Research (IISER), Research Intern

Mohali, India

June 2017 – Aug 2017

- Applied statistical methods to astrophysical data for modeling galactic mass distributions from radio observations.

Inter University Accelerator Centre (IUAC), Research Intern

New Delhi, India

June 2016

- Analyzed experimental time-of-flight data to study ionization processes in atomic collision systems.

Education

Ph.D. University of Hamburg, Physics - CGPA: 1.0, *magna cum laude*

Oct 2020 – April 2024

- Member of the [Helmholtz Data Science Graduate School \(DASHH\)](#) [↗](#).

- Advanced ptychography and quantitative imaging algorithms [↗](#),

M.Sc. University of Cologne, Physics - CGPA: 1.4

Oct 2017 – Jan 2019

- Member of [Collaborative Research Center \(CRC\) 183](#) [↗](#) -Entangled States of matter.
- Specialized in computational condensed matter physics and biophysics.

B.Sc. University of Delhi, Physics - Grade: 94.9%

2014 – 2017

(Hon.) • Executive Council Member, Physics and Electronics Society, St. Stephen's College.

Selected Publications *(full list on ORCID)*

- Interferometric measurement of nuclear resonant phase shift with a nanoscale Young double waveguide [↗](#), Nature Photonics (2026), *Co-author*
- Energy Time Ptychography for one-dimensional phase retrieval [↗](#), Optica (2025), *First author*
- Collective Nuclear Excitation and Pulse Propagation in Single-Mode X-Ray Waveguides [↗](#), Physical Review Letters (2025), *Co-author*

Awards

- 3rd place, [Helmholtz Data Science Challenge](#) [↗](#) - Machine learning on real world datasets 2022
- Bonn-Cologne Graduate School Scholarship 2018-2019
- IAS-INS-INS Summer Research Fellowship 2017
- INSPIRE fellowship, Department of Science and Technology, Government of India 2014-2017

Projects

Scientific Software Development

- Developed [NuPty](#) [↗](#), an open source Python/PyTorch library for inverse modeling and reconstruction of material properties from noisy, high-dimensional experimental data.
- Built modular scientific computing libraries for calibration and fitting workflows in non-linear X-ray splitting experiments, including [calibration tools](#) [↗](#) and [optimization routines](#) [↗](#). Used in multiple academic research projects (**1 Bachelor theses, 2 Master's theses, 1 PhD project**).

Computational Physics & Modeling

- Conceived and implemented a reconstruction framework for nuclear ptychography solving inverse problems in synchrotron spectroscopy, enabling high resolution Mössbauer material characterization. The method is intended for deployment at major international synchrotron facilities (DESY, ESRF, APS).
- Developed a numerical simulation model for non-linear crystallography to study crystal phase evolution under coupled X-ray and optical excitation.
- Built a domain adaptation pipeline for classification of microscopy images using CycleGAN-based augmentation and PCA-based feature perturbations, improving accuracy by **20%** and achieving **3rd place** in a data challenge. [Project repository](#) [↗](#)

High-Performance Computing & Numerical Simulation

- Processed multi-gigabyte experimental detector datasets using distributed computing on HPC clusters for scalable data reduction and analysis.
- Computed Lyapunov exponents for Hamiltonian models of topological insulators using transfer matrix methods across disorder and parameter sweeps, constructing phase diagrams and identifying phase transitions (Master's thesis work).